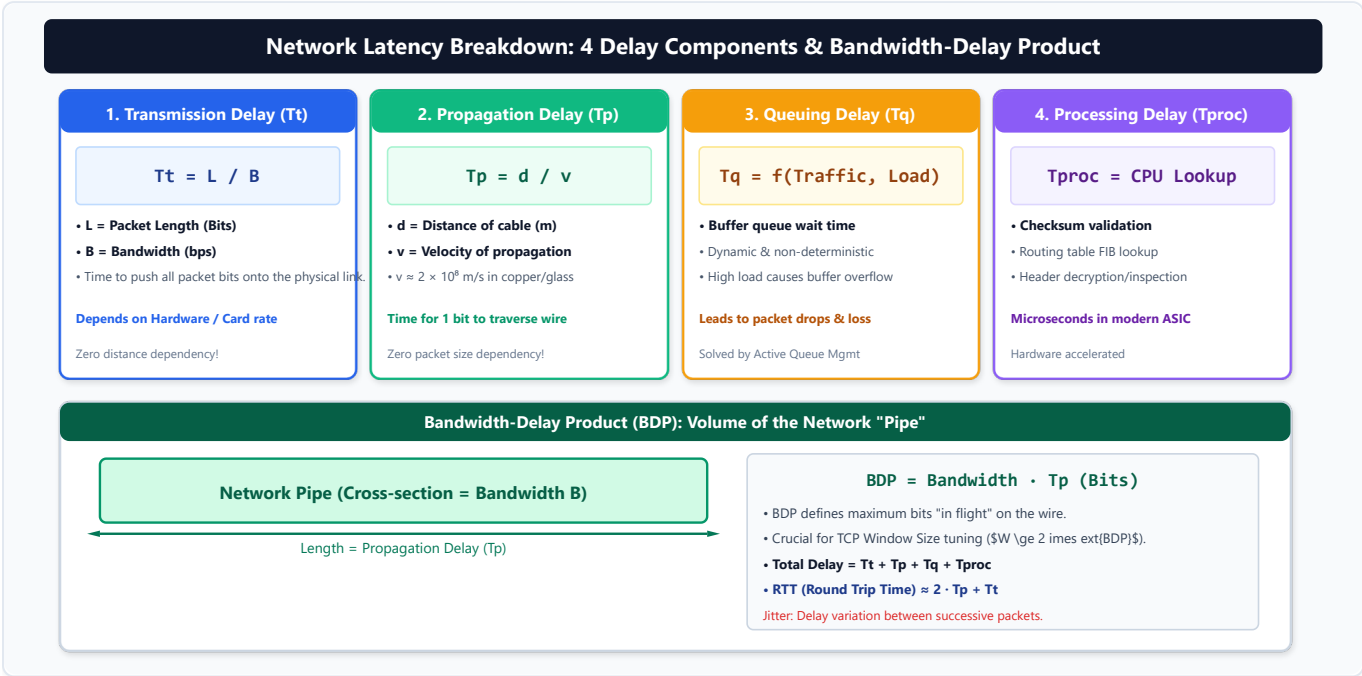


Module 10: Network Performance Metrics & Latency Analysis

AKTU BCS603 Focus: Latency decomposition (T_t , T_p , T_q , T_{proc}), Bandwidth-Delay Product (BDP), Round Trip Time (RTT), and jitter.



1. Transmission Delay vs Propagation Delay (AKTU Favorite)

Parameter	Transmission Delay (T_t)	Propagation Delay (T_p)
Core Formula	$T_t = L / B$	$T_p = d / v$
Governing Factors	Packet Size (L) & Link Bandwidth (B)	Cable Distance (d) & Wave Speed (v)
Analogous Meaning	Kitni der me pura packet car parking se nikla	Car ko highway par chalne me kitna time laga
Independent Of	Distance between nodes	Packet size and transmission bit rate

2. Channel Parameter Ratio α

Data link layer protocols ki efficiency ratio α par depend karti hai:

$$\alpha = \frac{T_p}{T_t} = \frac{d \cdot B}{v \cdot L}$$

Agar $\alpha \gg 1$ (High bandwidth, long distance jaise Satellite links), to Stop-and-Wait protocol extremely inefficient ho jata hai ($\approx \frac{1}{1 + 2\alpha} \approx 0$).

AKTU Exam Tip: Numerical solve karte waqt bytes ko hamesha pehle bits me convert karein ($1 \text{ ext{ Byte}} = 8 \text{ ext{ bits}}$), aur Mega (10^6) ya Kilo (10^3) units ko standard SI units me convert karna na bhoolein!